

that mirror and prism system, hence the name. Any camera without a DSLR style mirror system is technically speaking, a mirrorless camera, but we generally use this term to refer to more advanced cameras with interchangeable lenses.

- **DSLR:** If you want a camera that's flexible, fast and precise, a DSLR is what you're looking for. In this type of camera, the light enters through the lens and is bounced into an eye piece and focussing system by a mirror. When you press the trigger to release the shutter, the mirror and shutter both move out of the way to expose the sensor. Because of that dedicated focus sensor in particular, DSLRs are exceptionally fast at focussing on something, especially on moving objects.

Specifications that matter

Sensor size: This is by far the most important thing to consider when purchasing a camera. Put simply, the larger the sensor (the greater its area), the more data is captured and the better your image quality. Sensor size is the reason that a 12MP SLR captures better images than that 20MP phone camera.

The most prominent sensor sizes that matter are as follows:

- **Full-Frame:** This is the Holy Grail as far as the bulk of professional photographers are concerned. A full-frame sensor is one with a sensor size that is equivalent to that of a single 35mm film strip. This type of sensor is almost exclusively found in high-end DSLRs and is capable of offering the best image quality.
- **APS-C:** This is by far the most common sensor size in use today (barring point-n-shoot cameras) and unless you're a professional who knows better, you won't need more than this. In fact, for most of us, even the micro four-thirds mentioned below will be more than enough.
- **Micro four-thirds:** This is most commonly found in Olympus and Panasonic cameras, this sensor is smaller than that found on most interchangeable lens cameras (more on that later). The only real advantage of this sensor over the others is that given the small size of the sensor, lenses can be more compact and are cheaper to manufac-

ture, making this a very viable, portable alternative to a bulky DSLR.

- **Everything else:** Any sensor smaller than the ones mentioned above isn't really worth talking about. The size of a sensor is usually denoted by a single figure the same way that a monitor's size is given in inches. As with monitors, that figure is only the measure of the diagonal and not a measure of the sensor's aspect ratio or pixel density. Rule of thumb, the larger the better.

A word of caution, a diagonal unit is not necessarily accurate as far as image sensors go. Many manufacturers fudge these figures by using a different naming or measurement convention (the same way that a 27" CRT is actually smaller than a 27" LCD. The only sure measure of sensor size is the sensor area.

Type of sensor: Image sensors come in two prominent flavours, CCD and CMOS. Normally, you needn't bother with this, but there are certain scenarios where this is important.

- **CCD:** Most image sensors are CCD sensors and they're usually very good at capturing image data.
- **CMOS:** This is a very sensitive sensor that's cheap to manufacture, but expensive to fine-tune. Thus, you'll find this sensor on the cheapest of cameras and on more expensive models. Avoid the cheap cameras with CMOS sensors.

Aperture size: The purpose of a lens is to collimate the light rays entering it and the aperture determines the amount of collimation and the amount of light entering the camera system. As a rule of thumb a lens with a larger aperture is a better choice primarily because you're letting more light in and will thus capture cleaner images.

On camera lenses, aperture is determined by a figure such as f/1.8, f/4.5, etc. The smaller the number, the larger the aperture. Do bear in mind that on mobile phones and compact digital cameras this figure is largely irrelevant owing to the

HOW WE TEST

When testing cameras, we do all of the above but in environments with precisely controlled lighting with grey cards. We use a Minolta light meter to ensure that the luminance of a scene is exactly replicated from test to test and a SpyderCube to calibrate the white balance of a camera system.

When checking for tonal accuracy, each image is imported into Photoshop and we examine the colour of the SpyderCube in different lighting conditions. Since the SpyderCube is exactly 18% grey, the camera that captures a tone that's closest to the original gets the highest rating.

Regarding ergonomics, we examine just about every aspect of the camera and weightage is given to customization options, the presence of custom functions, accessibility of most frequently used features, etc. We also measure other things like the time to first photo (from an off state), time to focus under specific, controlled conditions, etc.

